pression in combustion chamber, efficient operation demands that all injector parts be maintained in clean, first-class condition at all times. This will necessitate the use of only clean fuel in injector, and intelligent care when servicing unit. No service operations should be attempted on injector without first acquiring a complete understanding of its construction and operation. This section of text thoroughly describes the working principle, shows the construction, and explains how to service unit.

While a skilled mechanic may intelligently service injectors, using the instructions in this section as a guide, time and expense may be saved by changing unit rather than attempting repairs especially in dirty surroundings.

Before starting to dismantle an injector, it is necessary to have an extremely clean workbench on which to work and to store parts. Cleanliness for the injector and its parts is emphasized because practically all injector service troubles are directly traceable to dirt, loose particles, grit, or other foreign matter.

When preparing to assemble an injector, it is absolutely essential to wash the hands thoroughly and to clean all injector parts in either clean fuel oil or carbon tetrachloride. Parts should then be blown dry by compressed air (filtered-free of dust and moisture), making sure to blow through all passages in injector body and all drilled holes, slots, etc., in injector parts. **Waste or rags should never be used for cleaning injector parts,** since this would leave lint and other particles which could collect and clog parts of injector when assembled.

Toilet tissue is a good and inexpensive material for wiping an injector part, or on which to place injector parts after cleaning with liquid and blowing dry with air. After liquid wash and subsequent drying with air, it is essential that parts be placed in a clean location until assembled. Clean, **light oil** should be used liberally on injector parts during assembly to protect them from rust in case injector is not used immediately and to provide initial lubrication when engine is started.

CARE OF INJECTOR (When Replacing)

Because of important part the injectors play in operation of engine, the necessity for proper care and cleanliness of these units cannot be overemphasized. Instructions below should be carefully followed:

1. Whenever the fuel lines are removed from an injector which is installed in engine, protect the two fuel fittings with shipping cap, part No. 5226414, to exclude dirt.

2. Do not dismantle an injector in dirty surroundings or on a dirty workbench. Use clean paper on the bench, and, after parts have been removed from injector, store loose parts in clean fuel oil as a protection against dirt and corrosion.

3. With more than one injector disassembled, it is not only good practice, but necessary, to keep parts of each injector separate. The plunger must be fitted, always with the same bushing. It is advisable to keep the parts of tip assembly spring, stop, spray tip valve and seat, as a unit, as it insures that "pop" pressure and calibration built into injector will remain essentially the same as when it was first tested and assembled.

4. After injectors have operated in an engine, the filter caps or filters should not be removed from injector when injector is in engine. If filter caps or filters are to be removed, injector must be completely disassembled and cleaned.

5. Whenever an injector has been removed and reinstalled, or a new injector installed in engine, the injector must be timed and racks positioned. Refer to "Control Rack Positioning" later in this section, also to "Timing Injector" in **ENGINE TUNE-UP** section of this manual.

6. Whenever an injector is to be out of service for an extended period, the injector should be purged as outlined under "Storing Injectors" later in this section.

7. Before installing any used or rebuilt injectors in an engine, test each injector. See "Injector Tests," later in this section.

INJECTOR REMOVAL

If it becomes necessary to remove one of the fuel injectors for inspection or replacement, follow the procedure given below:

1. Remove valve rocker cover.

2. Remove fuel lines from both the injector and the fuel connectors.

Immediately after removal of fuel lines from an injector, the two fuel feed fittings should be protected with shipping cap, part number 5226414, to prevent dirt entering injector.

3. If necessary, crank engine with the starter or a bar at flywheel ring gear until the three rocker arm